Real-World Data Project: Retail Sales Analysis

Domain: Retail | Dataset: Synthetic educational transaction data | Period: January–December 2025

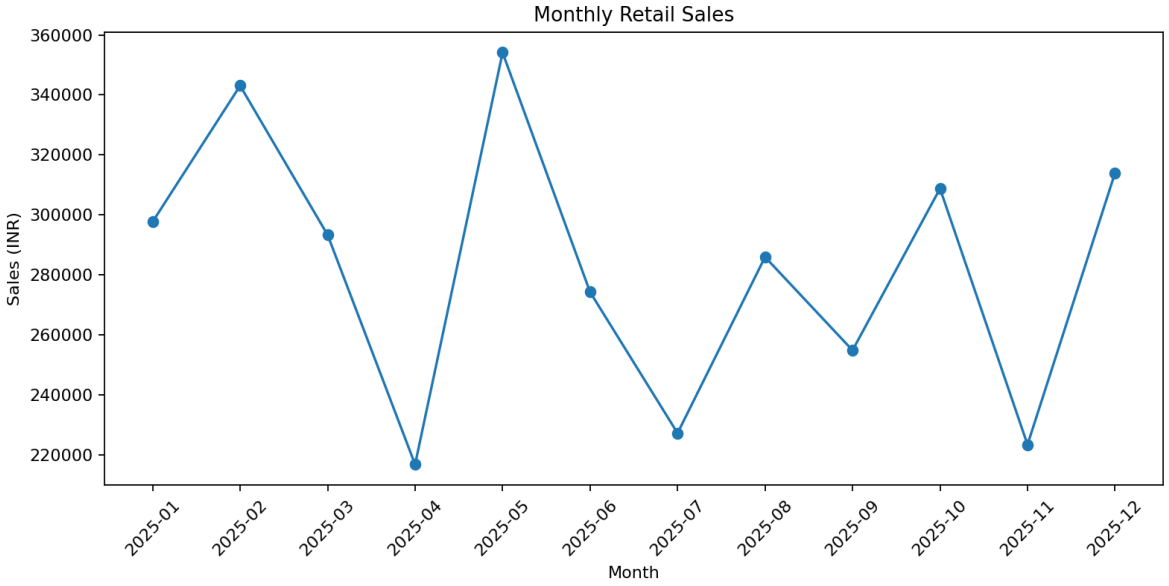
1. Introduction

This project applies data-analysis techniques to retail transaction data to identify sales patterns, profitability, regional performance, payment preferences, and top-performing products.

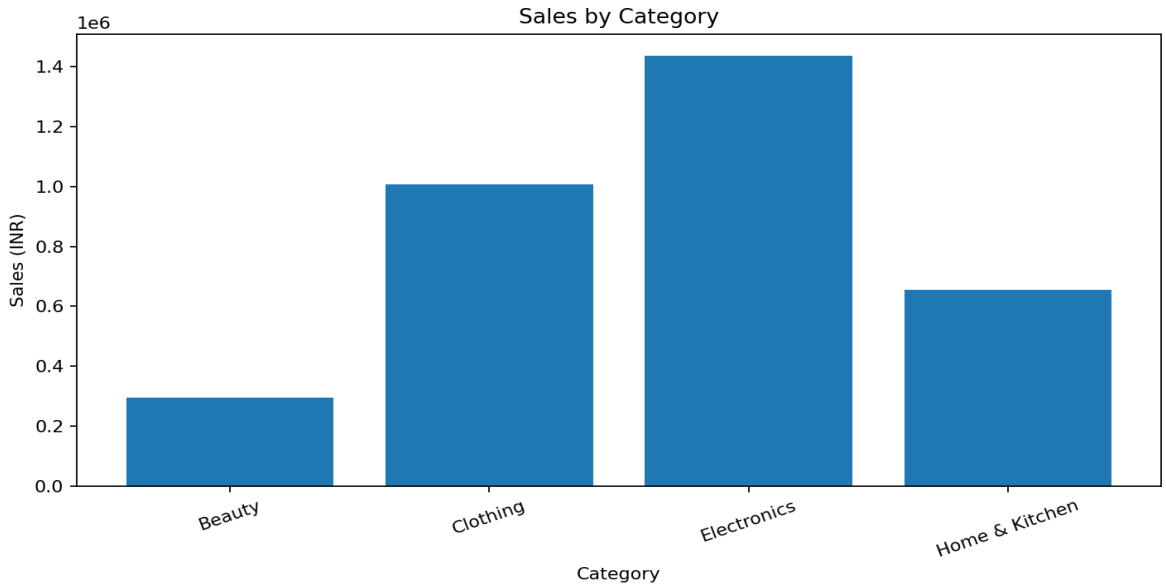
2. Key Performance Indicators

Metric	Result
Total Sales	■3,393,361.81
Total Profit	■1,039,314.65
Orders	1,200
Average Order Value	■2,827.80
Profit Margin	30.63%

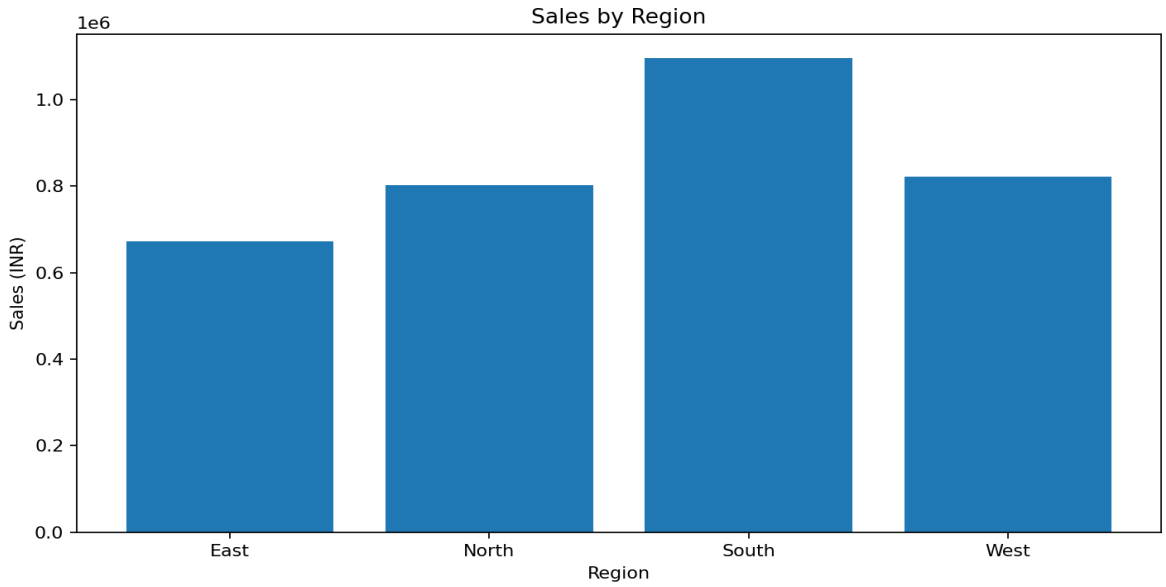
3. Monthly Sales Trend



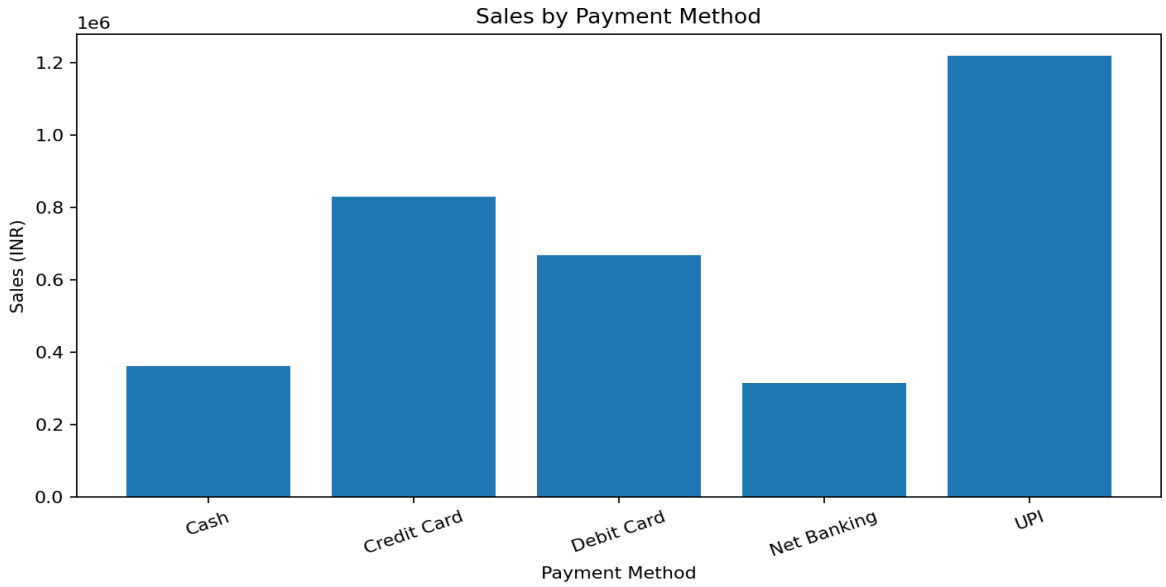
4. Sales by Category



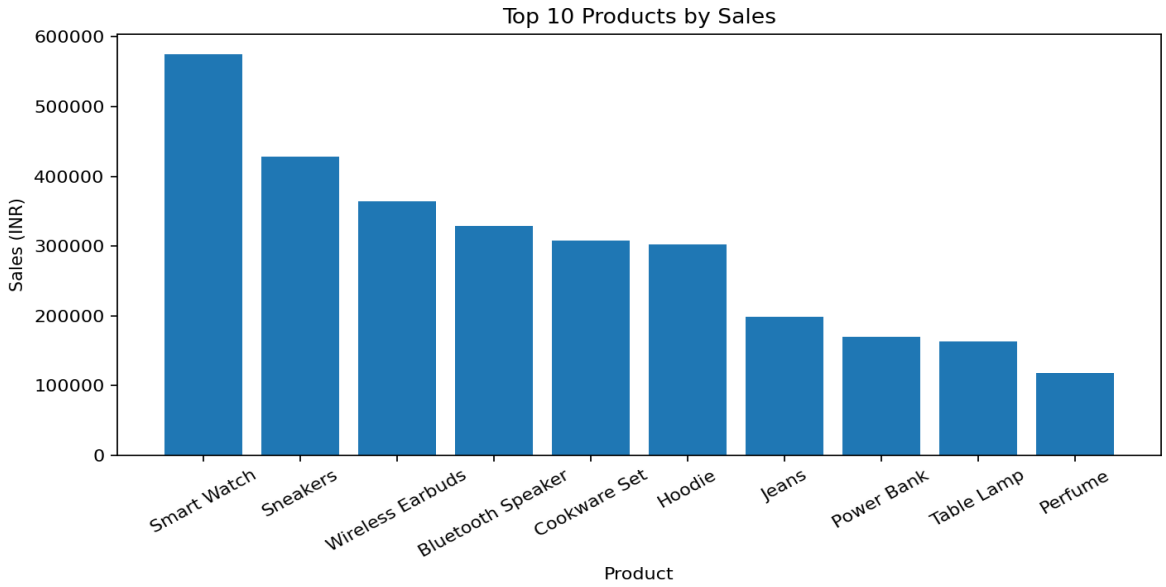
5. Sales by Region



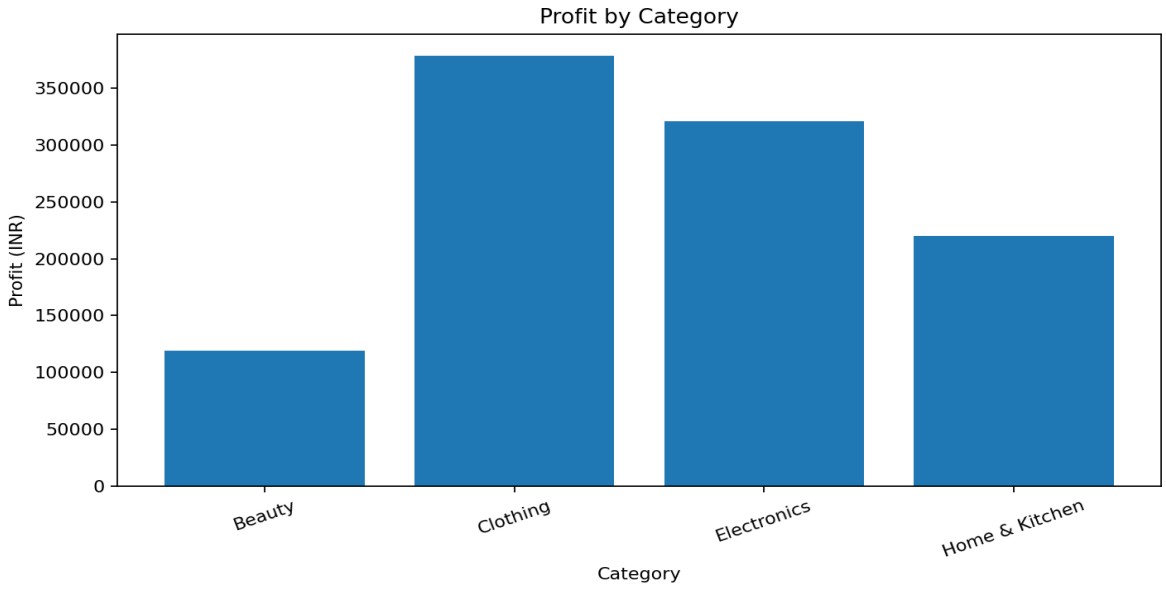
6. Sales by Payment Method



7. Top 10 Products



8. Profit by Category



9. Findings

• Electronics is the highest-sales category.
• South is the strongest region by sales.
• Smart Watch is the top product by sales.
• 2025-05 is the strongest month.
• Overall profit margin is approximately 30.63%.

10. Recommendations

Maintain inventory for high-performing products; use profitability as well as sales when planning promotions; strengthen marketing in high-performing regions while testing growth strategies elsewhere; support digital payment options; and review discounts to protect margins.

11. Conclusion

The project demonstrates an end-to-end retail analytics workflow: data preparation, KPI calculation, aggregation, visualization, interpretation, and business recommendations.